

SOC ALERT INVESTIGATION PORTFOLIO | AKACHI MADUAKO

SOC Alert Investigation Report

SOC Alert Investigation Portfolio - Mimikatz & Suspicious PowerShell

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This portfolio documents two related endpoint-security investigations using Microsoft Defender for Endpoint and Azure Log Analytics: a high-severity Mimikatz credential-dumping attempt and a medium-severity suspicious PowerShell/LaZagne retrieval alert.

1. Alert Overview

Alert Name	Mimikatz credential theft tool
Alert Category	Credential Access
Alert Type	Credential Dumping - Mimikatz
Severity	High
Classification	True Positive
Status / Outcome	Blocked and Remediated
Detection Source	Microsoft Defender for Endpoint / Antivirus
Affected Device	ec2amaz-9oh9aqq (Windows Server 2022)
User / Security Context	NT AUTHORITY\SYSTEM

2. Executive Summary

Microsoft Defender detected a high-severity Mimikatz credential theft alert on the affected Windows Server. The activity was associated with a suspicious process, wusvc.exe, running as SYSTEM. PowerShell attempted to download and execute Mimikatz with commands associated with credential dumping. Defender prevented execution of HackTool:Win32/Mimikatz.I and reported successful remediation. Based on the evidence reviewed, this alert is classified as a True Positive. The available evidence confirms an attempted credential-dumping operation, but does not by itself prove that credentials were successfully stolen. Sentinel/Log Analytics correlation also returned 16 Sysmon events involving wusvc.exe during the incident window, strengthening the process-chain evidence.

3. Key Evidence

- Suspicious process: C:\Users\Public\wusvc.exe (PID 4172).
- wusvc.exe executed with SYSTEM privileges and referenced http://172.31.2.203:8888.
- Microsoft Defender associated wusvc.exe with SandCat malware activity.
- PowerShell downloaded a Mimikatz archive from the GentilKiwi/Mimikatz GitHub repository.
- Mimikatz was invoked with privilege::debug and sekurlsa::logonpasswords, consistent with credential-dumping behavior.
- Defender detected HackTool:Win32/Mimikatz.I.
- Remediation action: remove; remediation result: Success.
- Alert status: Blocked.

4. Investigation Timeline

Time	Event	Analyst Interpretation
Aug. 16, 2026 - 2:45:11 PM	wusvc.exe observed in the process chain.	Suspicious malware-related process activity.
Aug. 16, 2026 - 2:45:16 PM	Defender detected SandCat-related	Supports malicious classification of

	malware activity.	the process chain.
Aug. 16, 2026 - 6:06:59 PM	Mimikatz execution attempt detected and blocked.	Credential-dumping attempt prevented by Defender.
Aug. 16, 2026 - 6:06:59 PM	Remediation action 'remove' completed successfully.	Endpoint protection successfully remediated the detected threat.

5. Indicators / Artifacts

Process	wusvc.exe
Process ID	4172
Process Path	C:\Users\Public\wusvc.exe
Observed Address	172.31.2.203:8888
Credential Tool	Mimikatz
Detected Threat	HackTool:Win32/Mimikatz.I
Related Malware	SandCat

6. MITRE ATT&CK Mapping

Primary mapping: Credential Access - OS Credential Dumping (T1003). The Mimikatz command shown in the evidence attempted to access logon credentials from memory. Add any additional technique mappings only when your investigation evidence clearly supports them.

7. Analyst Assessment

Classification: TRUE POSITIVE. The observed behavior is consistent with malicious credential-access activity. A suspicious SYSTEM-level process initiated PowerShell activity that attempted to download and execute Mimikatz. Microsoft Defender blocked the detected Mimikatz execution and successfully removed the threat. There is insufficient evidence in the reviewed alert to state that credentials were successfully dumped or exfiltrated.

8. Recommended Actions

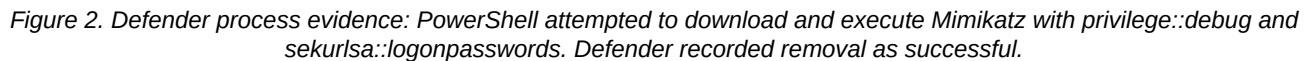
- Confirm the endpoint remains clean and review Defender for any related alerts on the same host.
- Investigate the origin and persistence mechanism of wusvc.exe and any related SandCat activity.
- Review activity involving 172.31.2.203:8888 and determine whether other hosts communicated with it.
- Review privileged account activity around the incident window for signs of credential misuse.
- Reset or rotate credentials if later evidence indicates credential exposure.
- Preserve relevant logs and document all containment and remediation actions.

9. KQL / Supporting Queries

Event

```
| where TimeGenerated between (datetime(2026-08-16 17:30:00) .. datetime(2026-08-16 19:00:00))
| where RenderedDescription contains "wusvc.exe"
| project TimeGenerated, EventID, Source, RenderedDescription
| order by TimeGenerated asc
```

Query purpose: Correlate the suspicious wusvc.exe process with Sysmon telemetry during the incident window. The query returned 16 events, including network connection (Event ID 3), process termination (Event ID 5), and process creation (Event ID 1) telemetry. The results include wusvc.exe PID 4172, matching the initiating process shown in Microsoft Defender.



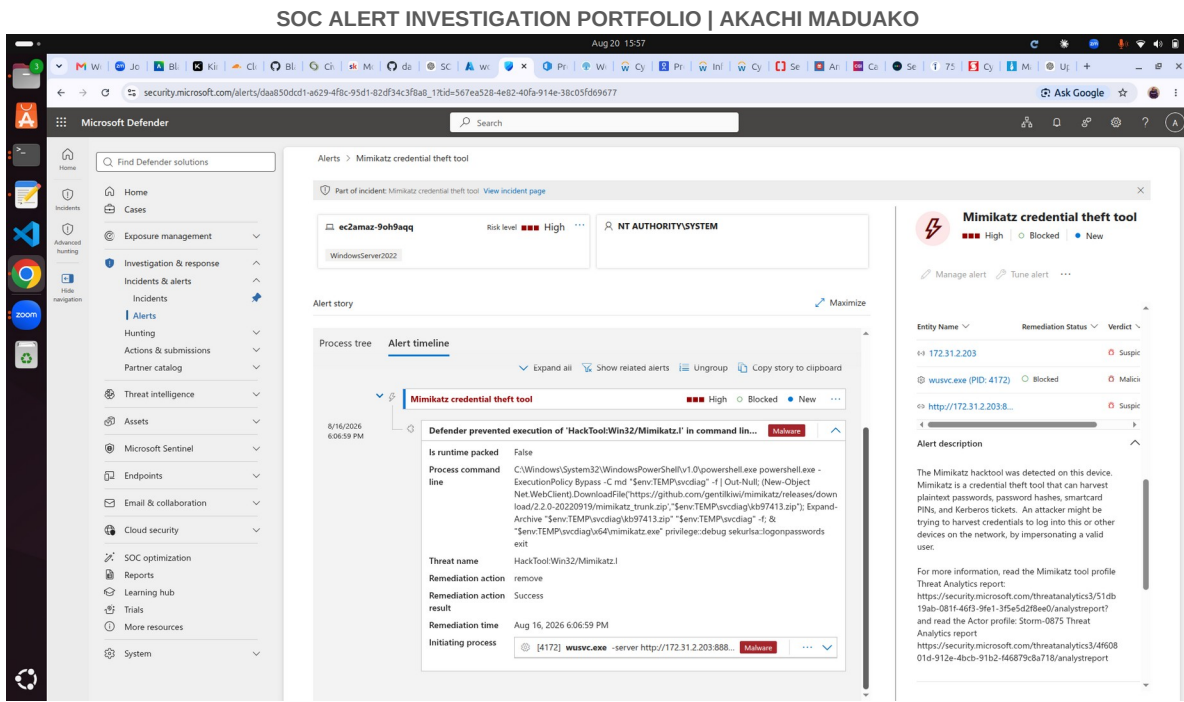


Figure 3. Defender alert timeline: HackTool:Win32/Mimikatz.I execution was prevented at 6:06:59 PM and linked to initiating process wusvc.exe PID 4172.

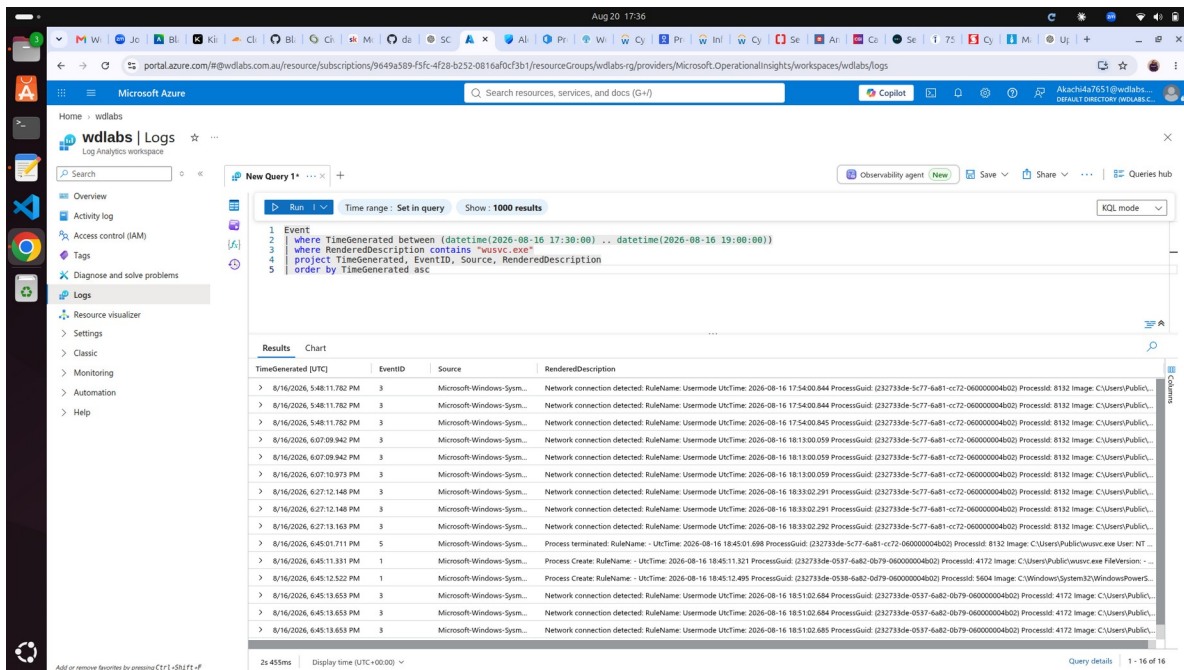


Figure 4. Sentinel/Log Analytics correlation: the wusvc.exe query returned 16 Sysmon events, including Event IDs 1, 3, and 5 around the incident window and PID 4172.

11. Lessons Learned / Analyst Notes

This investigation demonstrated how endpoint alert evidence can be validated with SIEM telemetry. Microsoft Defender provided the detection, process chain, command line, and remediation outcome, while Sentinel/Sysmon independently showed process and network activity involving wusvc.exe during the incident window. The most important analytical distinction was separating an attempted credential dump from confirmed credential theft: the evidence proves Mimikatz execution was attempted and blocked, but does not prove credentials were successfully obtained or exfiltrated.

12. Final Disposition

Final Classification	True Positive
Final Severity	High
Containment / Prevention	Mimikatz execution blocked by Microsoft Defender
Remediation	Threat removal reported successful
Credential Theft Confirmed?	No - attempted credential dumping is confirmed; successful theft is not established by the reviewed evidence.

ALERT 2 - SUSPICIOUS POWERSHELL COMMAND LINE

Alert Overview

- Alert Name: Suspicious PowerShell command line
- Severity: Medium
- Category: Execution
- Detection Source: EDR / Microsoft Defender for Endpoint
- Device: ec2amaz-9oh9aqq (Windows Server 2022)
- Security Context: NT AUTHORITY\SYSTEM
- Alert Status: New / Detected
- Incident: Execution incident on one endpoint
- First Activity: August 19, 2026 at approximately 2:10:53 AM
- Alert Generated: August 19, 2026 at approximately 2:12:04 AM

Executive Summary

Microsoft Defender for Endpoint generated a medium-severity alert after suspicious PowerShell activity was observed on ec2amaz-9oh9aqq. Defender evidence showed PowerShell running as NT AUTHORITY\SYSTEM with ExecutionPolicy Bypass and attempting to download LaZagne from its GitHub repository. LaZagne is a credential-recovery tool that can be abused to obtain stored credentials. The activity was treated as a true positive because the command behavior and credential-access tooling were inconsistent with normal administrative PowerShell use.

Key Evidence

- Process: powershell.exe
- Process ID shown in Defender: 5908
- User / security context: NT AUTHORITY\SYSTEM
- PowerShell option observed: -ExecutionPolicy Bypass
- Defender evidence showed a download from the AlessandroZ/LaZagne GitHub repository.
- Two related Defender alerts were visible: Suspicious PowerShell command line and Suspicious process executed PowerShell command.
- Defender categorized the alert as Execution and correlated it to one endpoint.

Investigation Timeline

- August 19, 2026, approximately 2:10:53 AM - suspicious PowerShell activity began according to Defender.
- PowerShell executed with ExecutionPolicy Bypass and attempted to retrieve LaZagne.
- August 19, 2026, approximately 2:12:04 AM - Defender generated the Suspicious PowerShell command line alert.
- Azure Log Analytics was queried around the attack window for PowerShell, LaZagne, and PID 5908.
- No matching Event-table record was found that allowed the Defender PowerShell process chain to be fully correlated in Log Analytics.

Indicators / Artifacts

- Executable: powershell.exe
- PID: 5908 (Defender evidence)
- Account: NT AUTHORITY\SYSTEM
- Command behavior: ExecutionPolicy Bypass
- External resource: GitHub repository associated with AlessandroZ/LaZagne
- Tool: LaZagne credential-recovery / credential-access utility

MITRE ATT&CK Mapping

- T1059.001 - Command and Scripting Interpreter: PowerShell
- T1105 - Ingress Tool Transfer (attempted retrieval of tooling from an external repository)
- T1555 - Credentials from Password Stores (relevant to LaZagne credential-recovery behavior)

Analyst Assessment

Classification: True Positive - Malicious/Suspicious Activity. Defender directly observed suspicious PowerShell execution and an attempt to obtain LaZagne. The use of ExecutionPolicy Bypass, SYSTEM context, and credential-access tooling materially increases confidence that the activity was not routine PowerShell administration. The corresponding detailed process telemetry could not be correlated in the Log Analytics Event table, so the parent process chain remains unconfirmed from that data source.

Recommended Actions

- Confirm the endpoint is contained or otherwise protected from further malicious activity.
- Review related Defender alerts and the incident timeline for additional execution or credential-access activity.
- Verify whether LaZagne or related files were created on disk and remove/quarantine malicious artifacts where present.
- Review credentials that may have been exposed and reset affected credentials where appropriate.
- Investigate the initiating/parent process in Defender telemetry if retained.
- Review why the Defender process telemetry was not available in the Log Analytics Event table and confirm expected endpoint logging/ingestion coverage.

KQL / Supporting Queries

```
Event
| where TimeGenerated between (datetime(2026-08-19 02:08:00) .. datetime(2026-08-19 02:13:00))
| where RenderedDescription contains "powershell"
| project TimeGenerated, EventID, Source, RenderedDescription
| order by TimeGenerated asc
```

```
search in (*) "LaZagne"
| where TimeGenerated between (datetime(2026-08-19 01:45:00) .. datetime(2026-08-19 02:30:00))
| order by TimeGenerated asc
```

```
Event
| where TimeGenerated between (datetime(2026-08-19 02:08:00) .. datetime(2026-08-19 02:13:00))
| where EventID == 1
| where RenderedDescription contains "5908"
| project TimeGenerated, RenderedDescription
| order by TimeGenerated asc
```

Evidence Screenshots

Selected screenshots from the Defender and Log Analytics investigation are included below.

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This screenshot shows the Microsoft Defender alert details for a 'Suspicious PowerShell command line' incident. The alert is classified as 'High' risk and is assigned to 'NT AUTHORITY\SYSTEM'. The alert details include the alert ID, detection source (EDR), detection status (Detected), and the first activity timestamp. The 'What happened' section provides a brief description of the suspicious PowerShell activity. The 'Recommended actions' section lists four steps for investigation, including examining the PowerShell command line, searching for indicators, exploring the timeline, and looking for related processes. The 'Process tree' and 'Alert timeline' sections are also visible, showing the process flow and the timeline of events.

Defender alert overview - Suspicious PowerShell command line.

This screenshot shows the Microsoft Defender alert details for a 'Suspicious PowerShell command line' incident, focusing on the process evidence and alert timeline. The 'Process tree' section shows the process flow, including the creation of the 'powershell.exe' process. The 'Alert timeline' section shows the timeline of events, including the execution of the PowerShell command. The 'Evidence' section shows the details of the detected process, including the entity name, remediation status, and verdict. The 'Alert description' section provides a detailed description of the suspicious PowerShell activity, including the command executed and the resulting output.

Defender process evidence showing PowerShell activity and the LaZagne repository reference.

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Microsoft Azure Log Analytics workspace showing a query for PowerShell events. The query filters for events where TimeGenerated is between 2026-08-19 01:45:00 and 2026-08-19 02:30:00, and the RenderedDescription contains 'powershell.exe'. The results table shows multiple entries for 'Process Create' events from Microsoft-Windows-Sysmon.

TimeGenerated [UTC]	EventID	Source	RenderedDescription
8/19/2026, 1:50:10.260 AM	1	Microsoft-Windows-Sysmon	Process Create RuleName: - UtcTime: 2026-08-19 01:50:10.260 ProcessGuid: (327733de-b4d7-4a85-5d95-060000004002) ProcessId: 7872 Image: C:\Windows\System32\WindowsPowerSh...
8/19/2026, 1:50:10.849 AM	11	Microsoft-Windows-Sysmon	File created RuleName: - UtcTime: 2026-08-19 01:50:10.829 ProcessGuid: (327733de-b4d7-4a85-5d95-060000004002) ProcessId: 7872 Image: C:\Windows\System32\WindowsPowerSh...
8/19/2026, 1:50:14.510 AM	1	Microsoft-Windows-Sysmon	Process Create RuleName: - UtcTime: 2026-08-19 01:50:14.506 ProcessGuid: (327733de-b4d7-4a85-5d95-060000004002) ProcessId: 6288 Image: C:\Windows\System32\WindowsPowerSh...
8/19/2026, 1:50:16.283 AM	11	Microsoft-Windows-Sysmon	File created RuleName: - UtcTime: 2026-08-19 01:50:16.259 ProcessGuid: (327733de-b4d7-4a85-5d95-060000004002) ProcessId: 6288 Image: C:\Windows\System32\WindowsPowerSh...
8/19/2026, 1:50:18.532 AM	1	Microsoft-Windows-Sysmon	Process Create RuleName: - UtcTime: 2026-08-19 01:53:08.506 ProcessGuid: (327733de-b4d7-4a85-5d95-060000004002) ProcessId: 8108 Image: C:\Windows\System32\WindowsPowerSh...
8/19/2026, 1:50:18.855 AM	11	Microsoft-Windows-Sysmon	File created RuleName: - UtcTime: 2026-08-19 01:53:08.835 ProcessGuid: (327733de-b4d7-4a85-5d95-060000004002) ProcessId: 8108 Image: C:\Windows\System32\WindowsPowerSh...
8/19/2026, 1:53:24.527 AM	11	Microsoft-Windows-Sysmon	File created RuleName: - UtcTime: 2026-08-19 01:53:24.525 ProcessGuid: (327733de-b4d7-4a85-5d95-060000004002) ProcessId: 8108 Image: C:\Windows\System32\WindowsPowerSh...
8/19/2026, 1:53:24.566 AM	1	Microsoft-Windows-Sysmon	Process Create RuleName: - UtcTime: 2026-08-19 01:53:24.557 ProcessGuid: (327733de-b4d7-4a85-5d95-060000004002) ProcessId: 5620 Image: C:\Windows\Microsoft.NET\Framework6...
8/19/2026, 1:53:25.118 AM	4109	Microsoft-Windows-PowerShell	CommandInvocationAdd-Type: - Add-Type: ParameterBindingAdd-Type: name="TypeDefinition" value=" using System; using System.Text; using System.Diagnostics; Tracing: using Mir...
8/19/2026, 2:01:58.796 AM	1	Microsoft-Windows-Sysmon	Process Create RuleName: - UtcTime: 2026-08-19 02:01:55.707 ProcessGuid: (327733de-b4d7-4a85-5d95-060000004002) ProcessId: 5600 Image: C:\Windows\System32\WindowsPowerSh...
8/19/2026, 2:01:58.855 AM	11	Microsoft-Windows-Sysmon	File created RuleName: - UtcTime: 2026-08-19 02:01:58.836 ProcessGuid: (327733de-b4d7-4a85-5d95-060000004002) ProcessId: 5600 Image: C:\Windows\System32\WindowsPowerSh...
8/19/2026, 2:02:29.559 AM	11	Microsoft-Windows-Sysmon	File created RuleName: - UtcTime: 2026-08-19 02:02:29.551 ProcessGuid: (327733de-b4d7-4a85-5d95-060000004002) ProcessId: 5600 Image: C:\Windows\System32\WindowsPowerSh...
8/19/2026, 2:02:29.559 AM	11	Microsoft-Windows-Sysmon	File created RuleName: - UtcTime: 2026-08-19 02:02:29.551 ProcessGuid: (327733de-b4d7-4a85-5d95-060000004002) ProcessId: 5600 Image: C:\Windows\System32\WindowsPowerSh...
8/19/2026, 2:02:29.594 AM	1	Microsoft-Windows-Sysmon	Process Create RuleName: - UtcTime: 2026-08-19 02:02:29.562 ProcessGuid: (327733de-b4d7-4a85-5d95-060000004002) ProcessId: 4884 Image: C:\Windows\Microsoft.NET\Framework6...

Log Analytics investigation of PowerShell-related events.

Microsoft Azure Log Analytics workspace showing a query for PID 5908 correlation. The query filters for events where TimeGenerated is between 2026-08-19 02:08:00 and 2026-08-19 02:13:00, and the RenderedDescription contains '5908'. The results table is empty, indicating no matching results.

TimeGenerated [UTC]	EventID	Source	RenderedDescription
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Final PID 5908 correlation query returned no matching Event-table result.

Lessons Learned / Analyst Notes

Endpoint alert evidence and Log Analytics telemetry may not contain identical fields or records. An investigation should preserve the Defender evidence while documenting any telemetry gap instead of assuming that a missing Event-table record invalidates the alert.

Final Disposition

- Verdict: True Positive - Malicious/Suspicious Activity
- Severity: Medium
- Primary Category: Execution
- Credential-access risk: Present due to attempted LaZagne retrieval
- Telemetry note: Defender evidence confirmed the activity; full process-chain correlation was not available in the Log Analytics Event table.